

Why Workers Stay: Pay, Beliefs, and Attachment

Sydnee Caldwell
UC Berkeley & NBER

Ingrid Haegele
LMU, CEPR & IZA

Joerg Heining
IAB

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Motivation

The willingness of workers to switch firms can affect what pay firms offer (Manning, 2003)

- ▶ $\eta = 3$: 10% wage cut leads to 30% of workers to leave (Sokolova and Sorensen, 2018)
- ▶ True in markets with many firms & in the “long-run”

Many reasons workers may not search or switch firms

- ▶ Beliefs about outside pay (average, or firm-specific)
- ▶ Preferences (including switching costs)

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What the key barrier to mobility is matters for policy & theory

- ▶ New vs. new classical monopsony
- ▶ Timing and efficacy of policy interventions (e.g. when to provide information)
- ▶ Beliefs about pay are at the core of directed vs. random search models

This Paper

1. What do workers believe about firm pay?
2. Do these beliefs affect workers' search behavior?
3. Why don't workers move?

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- ▶ Surveyed >10k full-time German workers through the IAB.
- ▶ Developed an infrastructure that allows us to link workers' answers about specific, named firms to data, incl. admin data on those firms.
- ▶ Asked workers what they think they would make at specific outside firms.

2. Do these beliefs affect workers' search behavior?

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- ▶ Asked workers whether they would apply to specific researcher-provided firms.

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- ▶ Asked workers how likely they would be to search if outside pay were X% higher.
- ▶ Firm-specific hypothetical choice experiments: rank hypothetical **offers** with randomized raises.
- ▶ Simulation: relative importance of beliefs and preferences.

Preview of Results

1. Workers do not believe in a uniform outside option.

- ▶ About half of workers report knowing wages at time of application.
- ▶ Workers expect their wages would vary across firms, holding position fixed.
- ▶ There are "firm effects" in workers' beliefs.
- ▶ Expected wages and wage premia are correlated with admin data predictions.

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- ▶ Joint test that workers direct their search and believe in firm rents.
- ▶ Amenity valuations are positively correlated with perceived firm wage premia.

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3. Many workers are not marginal to their current firm.

- ▶ Incumbent premium 7-18% of a worker's annual salary.
- ▶ Simulation: preferences better explain limited mobility.
- ▶ Amenity valuations differ between insiders and outsiders.

Related Literature

Competition in the Labor Market

- ▶ e.g. surveys in Sokolova and Sorensen (2018); Kline (2025); Caldwell et al. (2025)
- ▶ This paper: is limited mobility due to preferences or frictions?

Firm Wage Effects and Amenities

- ▶ e.g. Abowd et al. (1999); Card et al. (2013); Song et al. (2019); Rosen (1986); Sorkin (2018)
- ▶ This paper: workers expect firms to vary in wages and amenities, and believe $\text{cov}(a_j, \psi_j) > 0$

Workers' Information About the Labor Market

- ▶ e.g. Reynolds (1951); Caldwell and Harmon (2019); Jäger et al. (2024); Cullen (2023)
- ▶ This paper: a large share of workers have information on what specific outside firms would pay

Ethnographic Research on the Labor Market

- ▶ e.g. Myers and Shultz (1951); Reynolds (1951); Rees and Shultz (1970)
- ▶ This paper: compare workers' preferences and beliefs to admin. data

Framework

Model Sketch [See Paper]

Dynamic, partial-equilibrium on-the-job search model in which firms vary in wages and amenities

- ▶ Workers discount the future at rate δ , jobs are destroyed at rate λ

Workers search on and off the job and can apply to multiple firms

- ▶ Firms vary in offer probabilities p_k
- ▶ Pay per-firm application cost κ
- ▶ Switching cost s_i if accept an outside offer and move

Key feature: workers' flow utility depends on actual wages, but search depends on perceived wages

$$u_{ikt} = \beta w_{ikt} + a_k + \epsilon_{ikt} \text{ [actual]}$$

$$\tilde{u}_{ikt} = \beta \tilde{w}_{ikt} + a_k + \epsilon_{ikt} \text{ [perceived]}$$

Insights from Model [Formal Predictions in Paper]

Both Beliefs and Preferences Shape Search (and Therefore Mobility)

- ▶ Pessimistic pay beliefs deter productive search
- ▶ If switching costs are high, may not search even with overly optimistic beliefs
- ▶ If the covariance in amenities and perceived wages is negative (positive), will be less (more) responsive to pay

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Whether Workers' Pay Beliefs are Firm-Specific Matters

- ▶ Intuition: if cannot (can) map firm identities to different beliefs, need a random (the optimal) application to exceed search threshold
- ▶ Mistaken non-search is less common if workers can direct their search
- ▶ The firm-specific elasticity is increasing in the share of workers who direct their search

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Conditional on Offers, Mobility Depends on Preferences

- ▶ Influenced by both wage and non-wage factors, as well as switching costs

$$V_{ik} - s_i > V_{ij}$$

Data

Worker Survey and Linkages

Fielded a survey to full-time German workers between 25 and 50

- ▶ Invitations sent from the IAB by mail; survey completed online
- ▶ Initial survey (11.4% response rate) conducted in 2022 
- ▶ Follow-up survey (51% response rate) conducted in spring 2024 
- ▶ Randomized incentives to account for selection into non-response 
- ▶ Over-sampled workers at firms surveyed in Caldwell, Haegele and Heining (2024)

Linked the ~10k worker-level responses to IAB Social Security records

- ▶ Worker histories, occupation and industry codes, AKM effects (Bellmann et al., 2020)

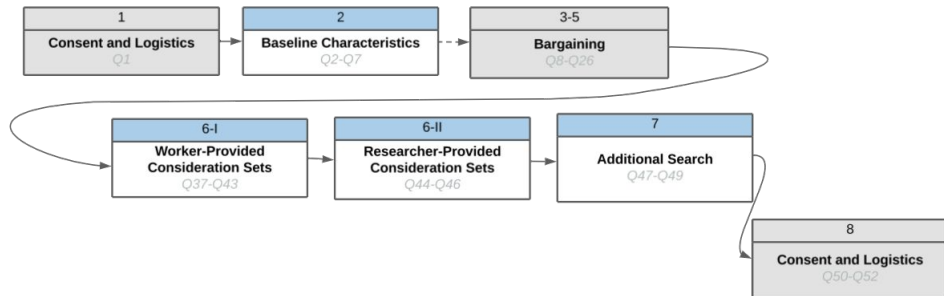
Constructed a novel linkage between workers' responses about specific firms and:

- ▶ Social Security records and wage premia associated with those firms
- ▶ ORBIS and Kununu 

Descriptive Statistics

	Unweighted		Initial Wave		Initial and Follow-Up	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Demographics</u>						
Female	0.31	(0.46)	0.40	(0.49)	0.37	(0.48)
Age	33.35	(6.23)	31.13	(5.18)	31.31	(5.19)
German Citizen	0.92	(0.27)	0.89	(0.32)	0.92	(0.28)
College Degree	0.60	(0.49)	0.53	(0.50)	0.61	(0.49)
Apprenticeship	0.33	(0.47)	0.37	(0.48)	0.31	(0.46)
<u>Employment</u>						
Daily Pay (Imputed)	180.05	(60.61)	143.42	(50.40)	150.77	(50.24)
Censored Pay	0.22	(0.41)	0.06	(0.24)	0.07	(0.25)
Hours (Survey)	40.27	(5.99)	40.36	(6.47)	40.43	(5.90)
CBA Covered (Survey)	0.59	(0.49)	0.48	(0.50)	0.45	(0.50)
Manufacturing Sector	0.47	(0.50)	0.22	(0.41)	0.23	(0.42)
Retail Sector	0.09	(0.28)	0.09	(0.29)	0.09	(0.29)
Professional Sector	0.13	(0.34)	0.15	(0.36)	0.17	(0.37)
Observations	9756		9756		3575	

Questions About Two Types of Firms



Selecting Researcher-Provided Firms

1. Not too many firms

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4. Horizontally reasonable 
5. Vertically reasonable

Selecting Researcher-Provided Firms

1. Not too many firms
 - ▶ 30 well-known German firms: 18 publicly listed, 12 family-owned.
 - ▶ Received more than 39.1 million page views on employer rating platform Kununu.
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 - ▶ 17 of the 30 firms are among the top 100 firms that workers named themselves.
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5. Vertically reasonable
 - ▶ Median and mean pay throughout observed worker distribution ▶

1. Eliciting Information About Search

Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.

- ☐ Company 1
- ☐ Company 2
- ☐ Company 3
- ☐ Company 4
- ☐ Company 5
- ☐ Company 6
- ☐ Company 7
- ☐ I would not apply to any of these

2. Eliciting Information About Wages

Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.

☐ Company 1

☐ Company 2

☐ Company 3

☐ Company 4

☐ Company 5

☐ Company 6

☐ Company 7

☐ I would not apply to any of these

What do you think your gross annual pay would be if you worked at these companies in a position similar to your current one?

Company 2: [Fill in gross pay]

Company 4: [Fill in gross pay]

Company 7: [Fill in gross pay]

3. Eliciting Preferences via Hypothetical Choice Experiments

Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.

- ☐ Company 1
- ☐ Company 2
- ☐ Company 3
- ☐ Company 4
- ☐ Company 5
- ☐ Company 6
- ☐ Company 7
- ☐ I would not apply to any of these

What do you think your gross annual pay would be if you worked at these companies in a position similar to your current one?

Company 2: [Fill in gross pay]

Company 4: [Fill in gross pay]

Company 7: [Fill in gross pay]

Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the raise specified.

Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Company 2 with a X% raise _____

Company 4 with a Y% raise _____

Company 7 with a Z% raise _____

Remain at current firm at current pay

Worker-Provided Firms

Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.

- ☐ Company 1
- ☐ Company 2
- ☐ Company 3
- ☐ Company 4
- ☐ Company 5
- ☐ Company 6
- ☐ Company 7
- ☐ I would not apply to any of these

“Suppose you planned to move to a new company in the next {one/three/six} months. What are companies that you would consider applying to?

Please list three companies that you would consider applying to and that hire employees in positions like yours (e.g. “Placeholder Inc”). These can be companies without current job vacancies.”

[Fill in Company 1]

[Fill in Company 2]

[Fill in Company 3]

☐ I do not want to answer this question

Worker-Provided Firms

- ▶ Worker-provided firms span a broad range of the German labor market
 - ▶ Total employment in Germany > 6.2 million.
 - ▶ Received more than 190 million page views on employer rating platform Kununu.
- ▶ Nearly 3000 distinct firms—of varying sizes—are mentioned 
 - ▶ Ex: local municipal utility, small manufacturer of agricultural equipment, glass producer

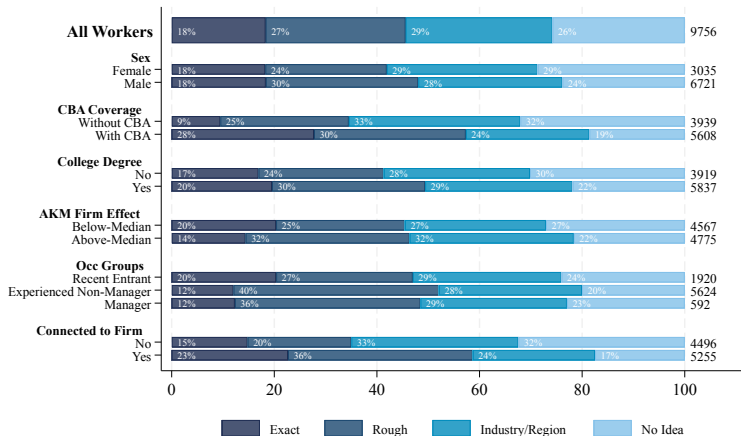
▶ Descriptive Statistics

▶ Comparison with Researcher-Provided

What Do Workers Know About Pay?

1. Many Workers Know Wages at the Time of Application

At the time that you applied, did you know what salary you would earn?



2. Workers Perceive There to Be A Firm Component to Pay

What do you think your gross annual salary would be if you worked at these companies in a position similar to your current one?

	Fraction Identical	Std. Deviation	Max/Min	N
	(1)	(2)	(3)	(4)
<u>A. Researcher-Provided Firms</u>				
Initial Survey	0.26	5673	1.18	3715
Follow-Up Survey	0.30	5294	1.15	3163
<u>B. Worker-Provided Firms</u>				
All Workers	0.25	5863	1.19	4433
All in Same State	0.22	4869	1.18	509
All in Same District	0.26	4701	1.19	173
All in Same Municipality	0.22	5084	1.21	159

Note: We asked 50% of workers to provide expected wages at specific researcher-provided firms. Column 1 reports the share of respondents who indicated the exact same expected pay across all three firms. Columns 2 and 3 report averages across respondents.

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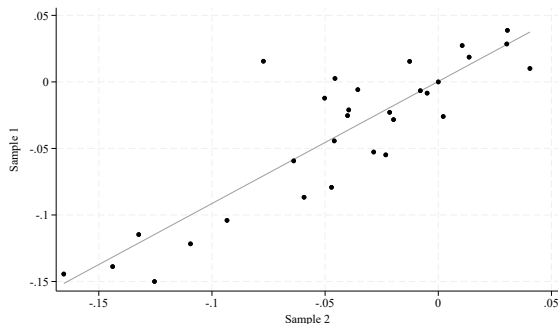
What do you think your gross annual salary would be if you worked at these companies in a position similar to your current one?

$$\log \tilde{w}_{ij} = \alpha_i + \psi_j + \epsilon_{it}. \quad (1)$$

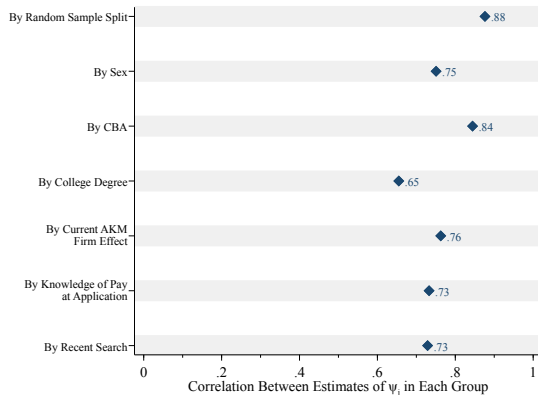
- ▶ Summarize workers' expectations at (randomly-chosen) researcher provided firms ▶ Balance
- ▶ Expected wage premia ψ_j
 - ▶ Identified using within-worker variation in expected salaries
 - ▶ Relative to a “base firm” whose premium is normalized to 0
- ▶ Worker fixed effects α_i
 - ▶ Portion of a worker's salary that they expect to be firm-invariant (across the provided firms)

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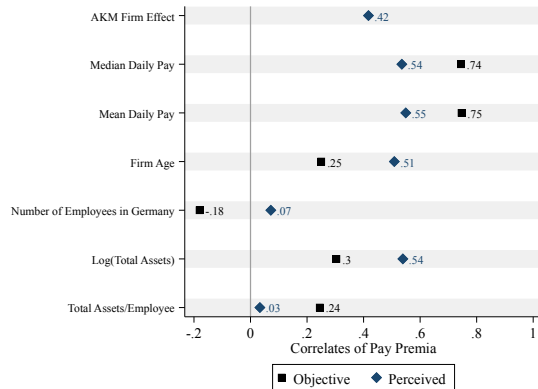
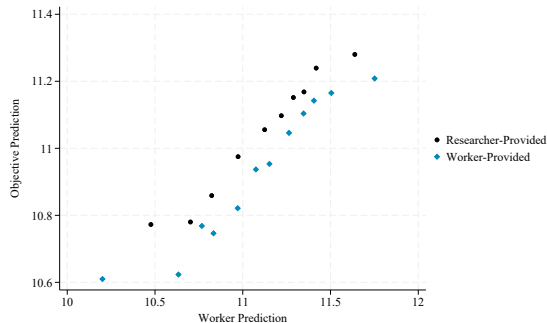
Split-Sample Estimates



Between-Group Correlation in ψ_j



3. Predictions are Correlated with Administrative Data Predictions



Pay and Search

Linking Search to Pay: Within-Worker, Within-Firm Design

$$\text{Consider } ij = \beta \log \tilde{w}_{ij} + X_{ij} + \gamma_j + \lambda_i + \gamma_{t(i)} + \epsilon_{ij}$$

	All Workers			Risk Tolerance		Would be Reluctant to Apply if P(Success) Were Low	
	(1)	(2)	(3)	Low	High	No	Yes
Own Pay Expectation	0.341*** (0.050)	0.313*** (0.050)	0.309*** (0.050)	0.256*** (0.056)	0.441*** (0.103)	0.297*** (0.084)	0.240*** (0.087)
Distance Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Same-Sector Control	No	No	Yes	Yes	Yes	Yes	Yes
Observations	21272	21272	21272	14967	6305	5392	6507
Number of Workers	6440	6440	6440	4519	1921	1476	1781
Test of equality (p-value)	---	---	---	.116		.64	
Mean of Outcome	0.311	0.311	0.311	0.310	0.313	0.274	0.305
Implied Elasticity	1.097*** (0.161)	1.007*** (0.161)	0.993*** (0.161)	0.827*** (0.182)	1.406*** (0.330)	1.082*** (0.307)	0.788*** (0.286)

Offer Probabilities are Relatively Less Important

	Consider the Amount of Competition		Believe High Wage Firms Get More Qualified Applicants	
	No	Yes	No	Yes
	(1)	(2)	(3)	(4)
Own-Pay Expectation	0.246*** (0.083)	0.271*** (0.083)	0.322*** (0.057)	0.228** (0.107)
Observations	6217	5685	16759	4513
Number of Workers (Clusters)	1713	1544	5190	1250
Test of equality (p-value)	0.830		0.439	
Mean of Outcome	0.283	0.300	0.312	0.304
Implied Elasticity	0.870*** (0.294)	0.905*** (0.277)	1.031*** (0.183)	0.752** (0.352)

Note: Regressions use data from researcher-provided firm module of initial and follow-up surveys.

Robustness: Alternative Designs

▶ Design 1: Link Consideration of Researcher-Provided Firms to:



1. Pay premia workers associate with that firm
2. Observed pay premia
3. Pay policy (Log of mean pay)

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► Design 2: Link Workers' Free-Text Responses to Pay Premia



Market Level Results

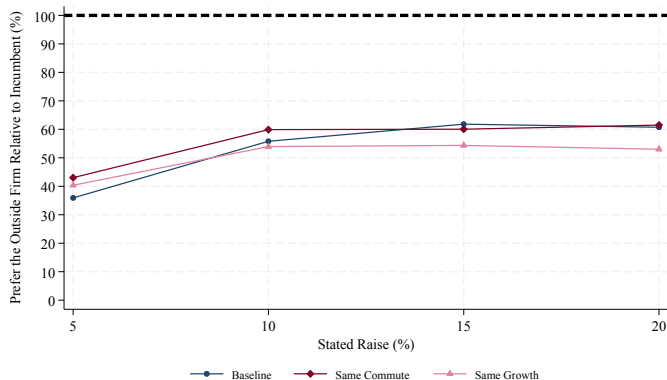
Imagine you were to discover that other companies in your area pay {X%} more than your current employer. How likely is it that you would start applying for jobs at other companies?

	All	Risk Tolerance		Would be Reluctant to Apply if P(Success) Were Low		Consider the Amount of Competition		Believe High Wage Firms Get More Applicants	
		Low	High	No	Yes	No	Yes	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Outside Pay	1.880*** (0.146)	1.847*** (0.172)	1.896*** (0.277)	2.207*** (0.216)	1.550*** (0.196)	1.702*** (0.216)	2.062*** (0.198)	1.932*** (0.180)	1.768*** (0.246)
Constant	0.263*** (0.020)	0.256*** (0.023)	0.289*** (0.040)	0.245*** (0.031)	0.286*** (0.026)	0.275*** (0.028)	0.252*** (0.028)	0.271*** (0.024)	0.251*** (0.033)
Implied Elasticity	7.139*** (1.058)	7.207*** (1.272)	6.559*** (1.818)	9.024*** (1.963)	5.423*** (1.143)	6.180*** (1.375)	8.188*** (1.640)	7.133*** (1.269)	7.048*** (1.866)
Observations	3557	2551	1006	1587	1954	1857	1685	2223	1334

Mobility

Not All Workers Want to Switch Firms

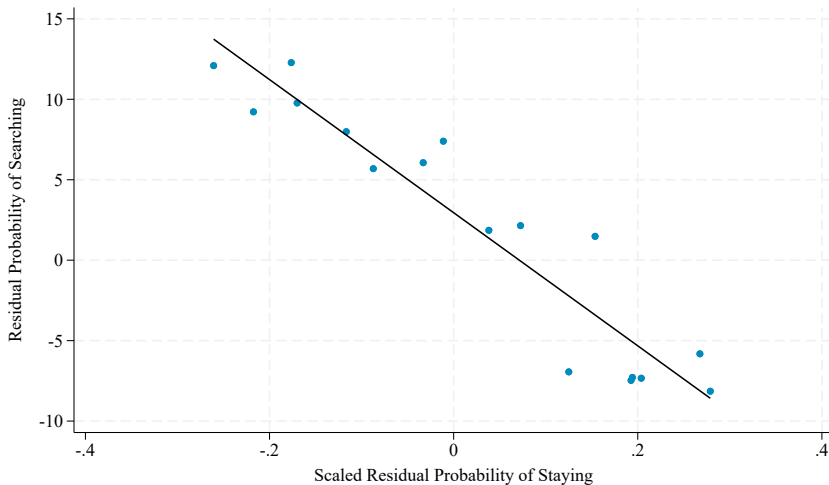
We asked workers to rank three outside firms with randomized raises and their inside firms with no raise.



Implied Labor Supply Elasticities

	Move to a Researcher- Provided Firm	Move to a Worker-Provided Firm		
		Baseline	Same Commute	Same Growth
	(1)	(2)	(3)	(4)
Outside Pay	0.391*** (0.133)	1.566*** (0.236)	1.288*** (0.364)	0.945** (0.368)
Constant	0.321*** (0.018)	0.359*** (0.033)	0.425*** (0.042)	0.408*** (0.043)
Implied Elasticity	1.219** (0.477)	4.364*** (1.037)	3.031*** (1.139)	2.320** (1.130)
Observations	21872	12217	6264	6272

Workers Who Do Not Want to Move Do Not Search



The Incumbent Premium

$$u_{ij} = \beta \log w_{ij} + c \cdot 1 \{j(i) \neq j\} + \epsilon_{ij} \quad (2)$$

	Move to a Researcher- Provided Firm		Move to a Worker-Provided Firm		
	Baseline	Distance Controls	Baseline	Same Commute	Same Growth
	(1)	(2)	(3)	(4)	(5)
Log Raise	6.172*** (0.492)	6.251*** (0.495)	8.112*** (0.824)	12.323*** (1.283)	10.207*** (1.207)
Incumbent	1.132*** (0.074)	0.703*** (0.129)	0.596*** (0.106)	0.771*** (0.134)	0.774*** (0.135)
Distance Controls	No	Yes	No	No	No
Observations	29961	29961	17539	8821	8782
Number of Workers	7735	7735	4796	2400	2385
Implied Incumbent Premium	0.183*** (0.009)	0.112*** (0.019)	0.074*** (0.008)	0.063*** (0.007)	0.076*** (0.008)

A Simple Simulation to Compare Beliefs and Preferences

- ▶ Assume workers hold compressed beliefs
 - ▶ If outside option is Δw , believe that it is $\lambda \Delta w$
 - ▶ Baseline: $\lambda = 0.5$ (conservative: beliefs are 25% as variable as observed)
 - ▶ Also consider $\lambda = 0.75$ and $\lambda = 0.25$

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 - ▶ Also consider $\lambda = 0.75$ and $\lambda = 0.25$
- ▶ Assign workers an outside option
 - ▶ Full population simulation: draw from the observed firm wage premium distribution
 - ▶ Below-median worker simulation: assign to the median wage-premium firm

A Simple Simulation to Compare Beliefs and Preferences

- ▶ Assume workers hold compressed beliefs
 - ▶ If outside option is Δw , believe that it is $\lambda \Delta w$
 - ▶ Baseline: $\lambda = 0.5$ (conservative: beliefs are 25% as variable as observed)
 - ▶ Also consider $\lambda = 0.75$ and $\lambda = 0.25$
- ▶ Assign workers an outside option
 - ▶ Full population simulation: draw from the observed firm wage premium distribution
 - ▶ Below-median worker simulation: assign to the median wage-premium firm
- ▶ Use empirical relationships between outside wages and search & mobility

$$\begin{aligned}\hat{\text{Search}}_i &= 1.88(\lambda \Delta w) + 0.263, \\ P(\text{Accept}) &= \frac{\exp(\beta \Delta w)}{\exp(\beta \Delta w) + \exp(\alpha)}\end{aligned}$$

Eliminating Attachment yields 3x the mobility of Full Transparency

	High Attachment ("Provided Firms")		Low Attachment ("Consideration Set Firms")	
	Search	Mobility	Search	Mobility
	(1)	(2)	(3)	(4)
A. Below Median-Wage Workers				
<u>Full Transparency</u>				
Baseline $\lambda = 0.5$	12%	7%	12%	9%
High Compression $\lambda = 0.25$	19%	10%	19%	13%
Low Compression $\lambda = 0.75$	6%	3%	6%	4%
<u>Eliminate Attachment</u>				
Baseline $\lambda = 0.5$	34%	24%	14%	12%

Eliminating Attachment yields 3x the mobility of Full Transparency

	High Attachment ("Provided Firms")		Low Attachment ("Consideration Set Firms")	
	Search	Mobility	Search	Mobility
	(1)	(2)	(3)	(4)
B. All Workers				
<u>Full Transparency</u>				
Baseline $\lambda = 0.5$	8%	6%	7%	7%
High Compression $\lambda = 0.25$	12%	9%	10%	10%
Low Compression $\lambda = 0.75$	4%	3%	3%	3%
<u>Eliminate Attachment</u>				
Baseline $\lambda = 0.5$	19%	15%	10%	8%

Unpacking Home Bias

Unpacking the Own-Firm Preference

$$u_{ij} = \beta \log w_{ij} + a_j + (\phi) \cdot 1 \{j(i) \neq j\} + \mu \log d_{ij} + \epsilon_{ij}$$

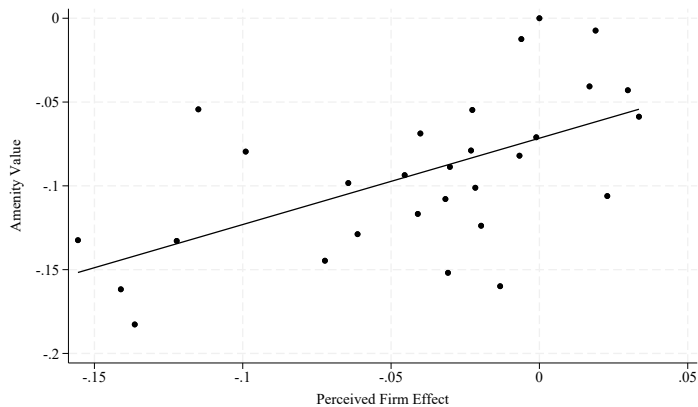
Use workers' preferences over provided firms with randomized raises to identify β and a_j

1. Do workers believe firms vary in ex ante rents ($a_j \neq 0$)
2. Do workers believe that higher wages are offset by lower amenities?
3. Do valuations differ between insiders and outsiders (as suggested by "personal connections")?

Firm-Specific Valuations of Amenities

	Preferences Over Outside Firms		Preferences Over All Firms		Preferences Over Considered or Incumbent Firm
	(1)	(2)	(3)	(4)	(5)
Log Raise (β)	8.560*** (3.090)	14.136*** (3.358)	6.162*** (2.322)	8.712*** (2.294)	15.647*** (4.409)
Observations	4217	4217	5671	5671	3001
Number of Workers (Clusters)	1177	1177	1200	1200	1192
Distance Controls	Yes	Yes	Yes	Yes	Yes
Data Include Incumbent Ranking	No	No	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Firm-Consider Interactions	No	Yes	No	Yes	No
Firm-Incumbent Interactions	No	No	Yes	Yes	Yes
<u>Amenities are Non-Zero</u> $a_j = 0 \forall j$					
p-value	<.001	<.001	<.001	<.001	<.001
Degrees of Freedom	29	29	29	29	29
<u>Consideration Doesn't Affect Valuations</u> $\zeta_j = 0 \forall j$					
p-value	---	<.001	---	<.001	---
Degrees of Freedom	---	30	---	30	---
<u>There is an Incumbent Effect</u> $\iota_j = 0 \forall j$					
p-value	---	---	<.001	<.001	<.001
Degrees of Freedom	---	---	18	18	17
<u>Insiders and Outsiders' Valuations Differ</u> $\iota_j = \iota_k \forall j, k$					
p-value	---	---	<.001	<.001	<.001
Degrees of Freedom	---	---	17	17	16
<u>Insiders and Outsiders Who Want to Move to the Firm Have Different Valuations</u> $\iota_j = \zeta_j \forall j$					
p-value	---	---	---	<.001	---
Degrees of Freedom	---	---	---	18	---

Firm-Specific Valuations of Amenities



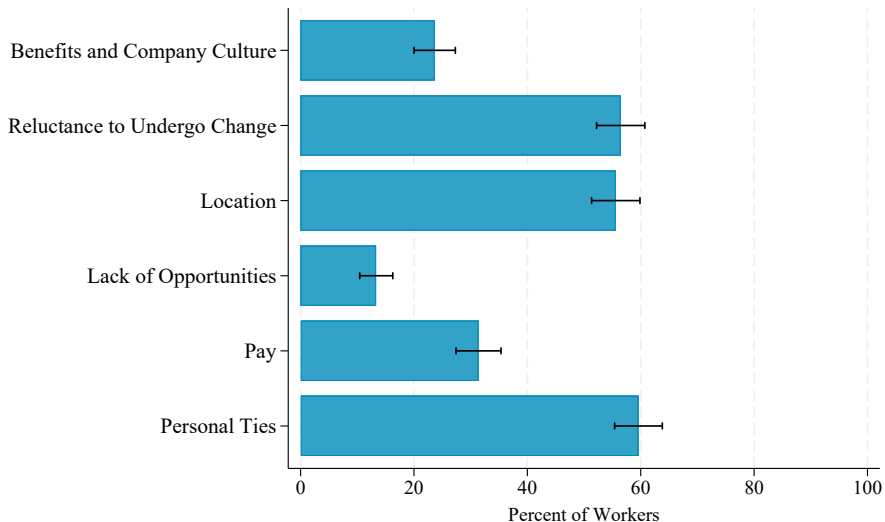
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	(1)	(2)	(3)	(4)	(5)
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Firm-Consider Interactions	No	Yes	No	Yes	No
Firm-Incumbent Interactions	No	No	Yes	Yes	Yes
<u>Amenities are Non-Zero</u> $a_j = 0 \forall j$					
p-value	<.001	<.001	<.001	<.001	<.001
Degrees of Freedom	29	29	29	29	29
<u>Consideration Doesn't Affect Valuations</u> $\zeta_j = 0 \forall j$					
p-value	---	<.001	---	<.001	---
Degrees of Freedom	---	30	---	30	---
<u>There is an Incumbent Effect</u> $\iota_j = 0 \forall j$					
p-value	---	---	<.001	<.001	<.001
Degrees of Freedom	---	---	18	18	17
<u>Insiders and Outsiders' Valuations Differ</u> $\iota_j = \iota_k \forall j, k$					
p-value	---	---	<.001	<.001	<.001
Degrees of Freedom	---	---	17	17	16
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p-value	---	---	---	<.001	---
Degrees of Freedom	---	---	---	18	---

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Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Firm-Consider Interactions	No	Yes	No	Yes	No
Firm-Incumbent Interactions	No	No	Yes	Yes	Yes
<u>Amenities are Non-Zero</u> $a_j = 0 \forall j$					
p-value	<.001	<.001	<.001	<.001	<.001
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p-value	---	<.001	---	<.001	---
Degrees of Freedom	---	30	---	30	---
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p-value	---	---	<.001	<.001	<.001
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p-value	---	---	---	<.001	---
Degrees of Freedom	---	---	---	18	---

Stated Reasons



Why Is There Limited Mobility?

- ▶ A large literature shows that there is limited mobility
- ▶ We show that this is likely due to **preferences** not information frictions
 - ▶ About half of workers say they had firm-specific pay when they joined their firm
 - ▶ Expectations are correlated with objective values
 - ▶ Labor supply elasticities of 2-4 based on randomized **offers**
 - ▶ Simulations suggest preferences explain more of limited mobility than beliefs
- ▶ Interesting direction for future work: Firms vary in ex post rents
 - ▶ Respondents suggest personal connections may be important

Appendix

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Selection

[▶ Back](#)

	Invited Mean	Linkage Consent			Panel and Linkage		Responded to Follow-Up		
		Mean	Difference		Mean	Difference	Mean	Difference Rel.	
			Rel. Invited			Rel. Linked		Invited	
	(1)	(2)	(3)		(4)	(5)	(6)	(7)	
<u>Demographics</u>									
Female	0.30 (0.46)	0.32 (0.46)	0.02 (0.00)	***	0.32 (0.47)	0.01 (0.01)		0.31 (0.46)	-0.01 (0.01)
Age	33.63 (6.59)	33.33 (6.23)	-0.32 (0.06)	***	33.33 (6.14)	-0.02 (0.17)		33.41 (6.16)	0.14 (0.14)
German Citizen	0.81 (0.39)	0.92 (0.27)	0.12 (0.00)	***	0.92 (0.26)	0.03 (0.01)	***	0.94 (0.24)	0.02 (0.01)
College Education	0.39 (0.49)	0.59 (0.49)	0.22 (0.01)	***	0.60 (0.49)	0.07 (0.01)	***	0.65 (0.48)	0.07 (0.01)
Apprenticeship	0.45 (0.50)	0.33 (0.47)	-0.12 (0.00)		0.32 (0.47)	-0.05 (0.01)		0.29 (0.45)	-0.06 (0.01)
Daily Earnings	146.03 (60.77)	169.79 (56.71)	25.69 (0.59)	***	170.92 (56.67)	6.61 (1.50)	***	175.02 (55.27)	7.27 (1.24)
Establishments	42705	3556			2983			1457	
Observations	134995	10134			8416			3664	

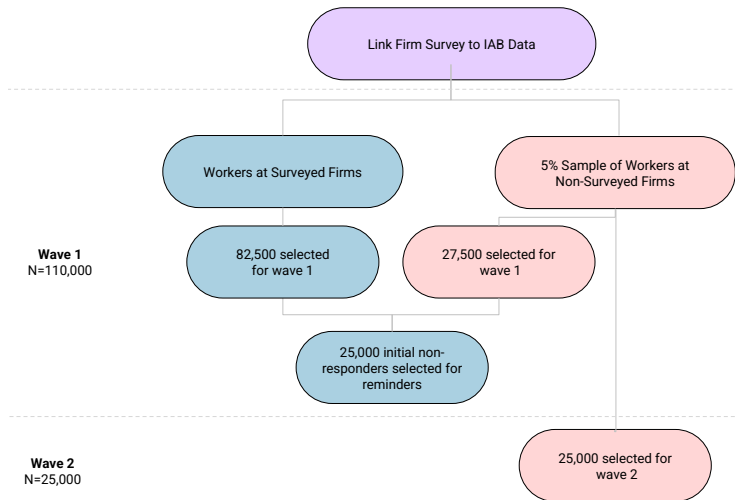
Incentives and Response (Initial Survey) [▶ Back](#)

	Endorsement	Gift Card		Reminder
	Letter	Level	Binary	
	(1)	(2)	(3)	(4)
Treatment	0.000	-0.000	-0.002	0.040***
	(0.002)	(0.000)	(0.002)	(0.001)
Observations	109995	109995	109995	99698

Incentives and Response (Follow-up Survey) [▶ Back](#)

	Provided an E-Mail			
	No	Yes		
	(1)	(2)	(3)	(4)
Reminder Letter	0.079*** (0.019)			
Initial Letter		0.230*** (0.015)		0.232*** (0.015)
Reminder E-mail			0.070*** (0.016)	0.077*** (0.016)
Constant	0.360*** (0.017)	0.270*** (0.013)	0.393*** (0.014)	0.210*** (0.018)
Observations	3405	5011	5011	5011

Worker Batches

[▶ Back](#)

**Deutsche Bahn AG**

als Arbeitgeber

auf kununu aktiv

[All und weitere](#)**3,8** ★★★★★**kununu Score**

13.153 Bewertungen

72%**Weiterempfehlung**

Letzte 2 Jahre

[Übersicht](#)[Bewertungen \(13.153\)](#)[Gehälter \(9.876\)](#)[Jobs \(579\)](#)[Firmenkultur \(3.281\)](#)[Fragen](#)[FOLGEN](#)[ARBEITGEBER BEWERTEN](#)

Mitarbeiter:innen-Zufriedenheit

▼ Karriere & Gehalt	3,3 ★★☆☆☆
▼ Unternehmenskultur	3,6 ★★☆☆☆
▼ Arbeitsumgebung	3,7 ★★☆☆☆
▼ Vielfalt	4,0 ★★★★★

Seit 2007 haben 13.153 Mitarbeiter und Bewerber diesen Arbeitgeber mit durchschnittlich 3,8 Punkten bewertet. Dieser Wert ist höher als der Durchschnitt der Branche [Transport/Verkehr/Logistik](#) (3,3 Punkte).

[Alle 13.153 Bewertungen entdecken](#)

Kenne deinen Wert und verhandle richtig.

9876 Deutsche Bahn Mitarbeiter haben auf kununu bereits für Gehaltstransparenz gesorgt. Finde heraus, wie viel du bei Deutsche Bahn verdienen kannst.

Sortierung: **Anzahl Gehaltsangaben** ▾**Lokomotivführer:in**Ø **40.800 €** brutto/Jahr

704 Gehaltsangaben

**Projektmanager:in**Ø **64.600 €** brutto/Jahr

700 Gehaltsangaben

**Fahrdienstleiter:in**Ø **40.900 €** brutto/Jahr

474 Gehaltsangaben



Gehaltszufriedenheit

**56%****sind mit ihren Gehältern
zufrieden** (basierend auf 8.575
Bewertungen)**Gehalt & Sozialleistungen** **3,5** ★★★★★

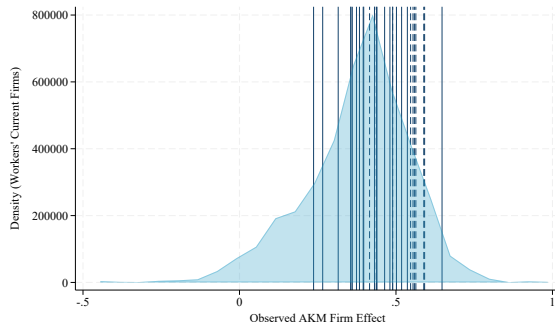
Basierend auf 8.575 Bewertungen

Gehaltsinformationen

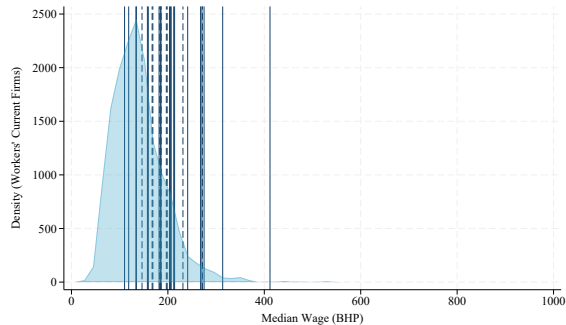
Aus dem Gehalt wird allzu gern ein Geheimnis gemacht. Gemeinsam mit dir und Millionen anderen kununu Usern können wir für notwendige Gehaltstransparenz sorgen. So findest du den Arbeitgeber, der wirklich zu dir und deinen Vorstellungen passt.

Researcher-Provided Firms vs. Workers' Current Firms [▶ Back](#)

AKM Firm Effect



Median Pay



Knowledge of Wages at the Time of Application: Question [▶ Back](#)

At the time that you applied, did you know what salary you would earn?

- ▶ I had no or very little idea
- ▶ I only had a rough idea what is paid in my region or sector
- ▶ I had at least a rough idea what this company pays for the position
- ▶ I knew exactly what this company pays for the position

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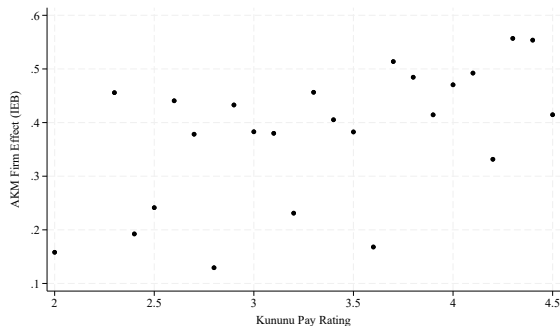
Correlates of Perceived AKM Effects [▶ Back](#)

	Researcher-Provided		Worker-Provided Firms	
	Firms			
	Observed Ψ	Estimated Ψ	Observed Ψ	Estimated Ψ
	(1)	(2)	(3)	(4)
<u>Firm Pay</u>				
Log(Mean Daily Pay) (BHP)	0.57	0.55	0.50	0.27
Log(Median Daily Pay) (BHP)	0.56	0.53	0.48	0.26
AKM FE	1.00	0.33	1.00	0.15
<u>Firm Characteristics</u>				
Age	0.31	0.27	0.01	0.01
Log(Overall Employees)	-0.13	0.42	0.53	0.08
Log(Employees in Germany)	0.03	0.32	-0.06	0.09
Log(Total Assets)	0.30	0.53	0.18	0.21
Log(Fixed Assets)	0.30	0.61	0.13	0.21
<u>Employer Ratings (Kununu)</u>				
Number of reviews	-0.06	-0.21	0.05	0.08
Pct. that would recommend	0.09	0.40	0.14	0.11
Salary rating	0.36	0.52	0.28	0.18
Top salary rating	0.09	0.40	0.14	0.11
	30		715	

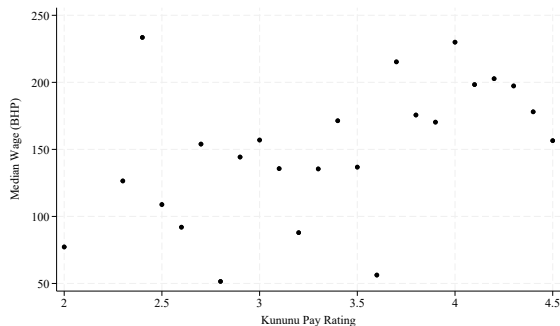
Quality of Online Salary Data

[▶ Back](#)

AKM Firm Effect



Median Pay



Note: Both binned scatterplots are weighted by firm size.

W

Comparison with AKM Estimates [▶ Back](#)

	Worker Expectations					Objective Predictions
	All Workers	Informed at Application		Recent Search Activity		
		Yes	No	Yes	No	
(1)	(2)	(3)	(4)	(5)	(6)	
<u>Number of Parameters</u>						
Person Effects	5305	2643	2662	3971	1334	1103
Firm Effects	30	30	30	30	30	29
<u>Summary of Parameter Estimates</u>						
Std. Dev. Person Effects	0.365	0.386	0.344	0.358	0.386	0.516
Std. Dev. Firm Effects	0.051	0.055	0.049	0.051	0.053	0.091
RMSE	0.105	0.108	0.102	0.105	0.106	0.315
<u>Addendum</u>						
Std. Dev. Log(Salary)	0.378	0.401	0.357	0.371	0.401	0.254
Variance Log(Salary)	0.143	0.160	0.127	0.138	0.160	0.064
Observations	19431	9692	9739	14580	4851	3879

Agreement in Estimates Across Demographic Groups [▶ Back](#)

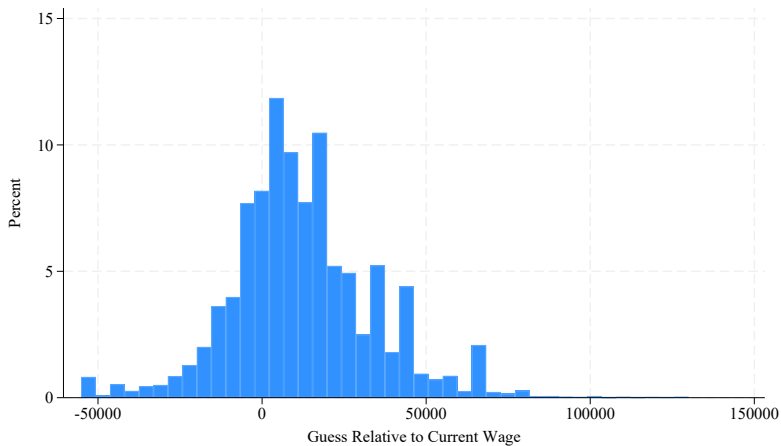
	Baseline Model		Rank-Ordered Logit	
	Correlation	Test of Equality (p-value)	Correlation	Test of Equality (p-value)
	(1)	(2)	(3)	(4)
Split-Sample	0.88	0.85	0.94	0.25
Sex	0.75	0.02	0.87	0.81
CBA	0.84	0.19	0.91	0.84
College Education	0.65	0.12	0.90	0.52
Current Firm AKM Effect (Split at Median)	0.76	0.01	0.90	0.33
Searched in Past 6 Mo.	0.73	0.02	0.86	0.73
Knew Wages at Application	0.73	0.20	0.92	0.56
Easy to Get a Better Job	0.82	0.38	0.93	0.93
Tenure (Split at 2 Years)	0.86	0.47	0.94	0.45

Firm Effects on Expectations are Non-Zero [▶ Back](#)

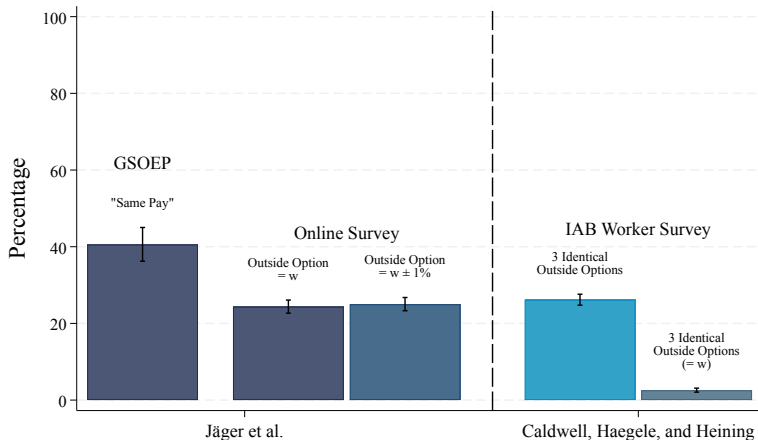
	Sector Fixed Effects		Sector and State Fixed Effects		Sector-State Fixed Effects	
	(1)	(2)	(3)	(4)	(5)	(6)
A. Researcher-Provided Firms						
Adjusted R-Squared	0.858	0.866	0.862	0.866	---	---
Observations	19734	19734	19734	19734	---	---
Parameters Tested		24		17	---	---
F-Statistic		7		5		
p-value		<.01		<.01	---	---
B. Worker-Provided Firms						
Adjusted R-Squared	0.888	0.907	0.868	0.885	0.871	0.881
Observations	10733	10733	8527	8527	8138	8138
Parameters Tested		535		387		334
F-Statistic		17370894		392000000000		18485608
p-value		<.01		<.01		<.01

Note: This table reports the adjusted R-squared from regressions of log expected earnings at researcher-provided (Panel A) or worker-provided (Panel B) firms. Columns 1, 3, and 5 include worker fixed effects, and the fixed effects indicated in the column. Columns 2, 4, and 6 add firm fixed effects. The F-statistic tests whether firm dummies are jointly zero. Standard errors are clustered at the worker level. Regressions use sampling weights. We exclude results with limited within-sector/state firm pairs.

Anchoring of Workers' Beliefs

[▶ Back](#)[▶ Firm Fixed Effects are Non-Zero](#)[▶ Comparison](#)

Comparison of Anchoring with Jaeger et al. (2025) [▶ Back](#)



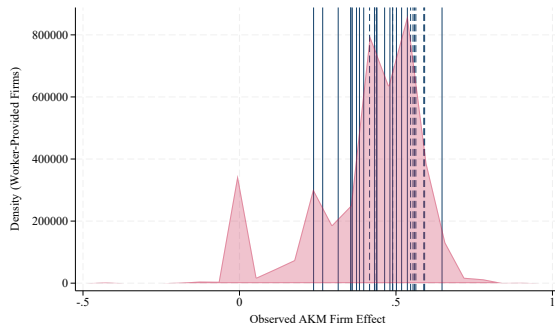
Number of Distinct Firms Mentioned by Workers [▶ Back](#)

	Firms by Number of Mentions		
	All	Linked to IEB	Linked to Orbis
	(1)	(2)	(3)
1 Time	1979	1302	1926
2-9 Times	719	500	645
10-49 Times	111	89	106
50-99 Times	13	11	13
100-249 Times	12	9	11
250+ Times	8	6	7
Total	2842	1917	2708

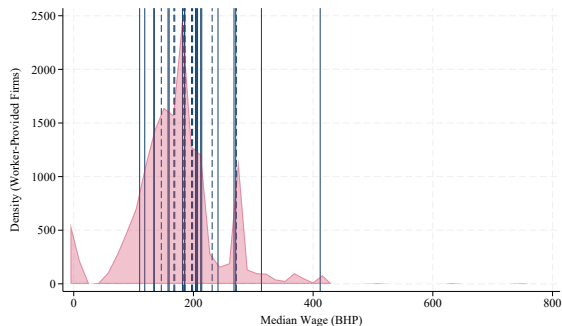
Observed Characteristics of Worker-Provided Firms [▶ Back](#)

Researcher-Provided Firms vs. Worker-Provided Firms [▶ Back](#)

AKM Firm Effect



Median Pay

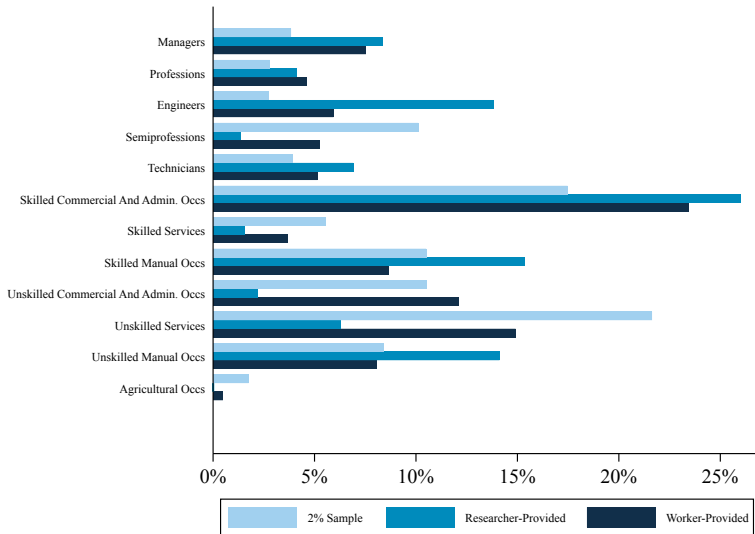


Characteristics of Researcher-Provided Firms [▶ Back](#)

Researcher-Provided vs Worker-Provided Firms

[▶ Back](#)

Occupational Distribution

[▶ Back](#)

Randomization of Firms to Workers [▶ Back](#)

	Initial Survey			Follow-Up		
	Firm Group	Firm	Firm Quality	Firm Group	Firm	Firm Quality
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Demographics</u>						
Female	0.375	0.576	0.570	0.094	0.064	0.570
Age	0.094	0.332	0.817	0.886	0.783	0.817
German Citizen	0.334	0.374	0.195	0.591	0.697	0.195
<u>Education</u>						
College	0.526	0.643	0.947	0.005	0.014	0.947
Apprenticeship	0.496	0.714	0.607	0.069	0.146	0.607
<u>Employment and Earnings</u>						
Daily Earnings	0.228	0.137	0.552	0.941	0.971	0.552
Earnings are Censored	0.391	0.764	0.893	0.682	0.822	0.893
Weekly Hours (Survey)	0.085	0.128	0.106	0.451	0.639	0.106
Covered by a CBA (Survey)	0.351	0.785	0.736	0.882	0.965	0.736
<u>Sector</u>						
Manufacturing	0.998	0.999	0.941	0.481	0.704	0.941
Retail	0.628	0.945	0.297	0.813	0.818	0.297
Professional	0.730	0.980	0.785	0.399	0.360	0.785

Note: Firm group is an indicator for the set of firms a worker is assigned to, firm is an indicator for the firm a worker is assigned to, and firm quality is the objective pay premium (Bellmann et al., 2020). We perform separate regressions of each covariate (indicated in the row) on the characteristics indicated in the column. Each entry provides the p-value from an F-test that all of the included regressor(s) (other than the constant) are equal to zero. P-values are calculated using standard errors clustered at the worker level. We assigned each worker to one of 18 groups, which defined the set of firms they would see.

Linking Search to Pay: Researcher-Provided Firms [▶ Back](#)

$$\text{Consider}_{ij} = \beta^{RP} \text{Pay}_j + \lambda_i + X_{ij} + \epsilon_{ij}$$

	Stated Consideration			Free-Text Responses		
	(1)	(2)	(3)	(4)	(5)	(6)
Mean of Dependent Variable		0.254			0.017	
A. Perceived Firm Effect (Split-Sample)						
Firm Premium (Split-Sample)	0.892*** (0.110)	0.983*** (0.127)	0.986*** (0.127)	0.097*** (0.011)	0.099*** (0.013)	0.101*** (0.013)
Observations	89742	89742	89742	224388	224388	224388
Number of Workers	9756	9756	9756	9756	9756	9756
B. Observed Firm Effect						
Firm Premium (Observed)	0.174*** (0.033)	0.165*** (0.033)	0.173*** (0.035)	0.014*** (0.005)	0.014*** (0.005)	0.014*** (0.005)
Observations	89258	89258	89258	214632	214632	214632
Number of Workers	9756	9756	9756	9756	9756	9756
C. Observed Log(Mean Daily Pay)						
Firm Mean Daily Pay	0.086*** (0.011)	0.093*** (0.011)	0.093*** (0.011)	0.008*** (0.002)	0.008*** (0.002)	0.008*** (0.002)
Observations	89258	89258	89258	214632	214632	214632
Number of Workers	9756	9756	9756	9756	9756	9756
Firm Characteristics						
	Size	Size, Brand Recognition	Size, Brand Recognition, CBA	Size	Size, Brand Recognition	Size, Brand Recognition, CBA
Fixed Effects	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector

Note: The outcome variable in Columns 1-3 is an indicator for whether the worker checked the box indicating they would consider applying to the firm if they wanted to switch firms. Regressions use sampling weights. Standard errors are clustered at the worker level.

Linking Search to Pay: Worker-Provided Firms [▶ Back](#)

$$\text{List Firm}_{ij} = \beta^{WP} \text{Pay}_j + \lambda_i + X_{ij} + \epsilon_{ij}$$

	Stated Consideration			Free-Text Responses		
	(1)	(2)	(3)	(4)	(5)	(6)
Mean of Dependent Variable		0.254			0.017	
A. Perceived Firm Effect (Split-Sample)						
Firm Premium (Split-Sample)	0.892*** (0.110)	0.983*** (0.127)	0.986*** (0.127)	0.097*** (0.011)	0.099*** (0.013)	0.101*** (0.013)
Observations	89742	89742	89742	224388	224388	224388
Number of Workers	9756	9756	9756	9756	9756	9756
B. Observed Firm Effect						
Firm Premium (Observed)	0.174*** (0.033)	0.165*** (0.033)	0.173*** (0.035)	0.014*** (0.005)	0.014*** (0.005)	0.014*** (0.005)
Observations	89258	89258	89258	214632	214632	214632
Number of Workers	9756	9756	9756	9756	9756	9756
C. Observed Log(Mean Daily Pay)						
Firm Mean Daily Pay	0.086*** (0.011)	0.093*** (0.011)	0.093*** (0.011)	0.008*** (0.002)	0.008*** (0.002)	0.008*** (0.002)
Observations	89258	89258	89258	214632	214632	214632
Number of Workers	9756	9756	9756	9756	9756	9756
Firm Characteristics						
	Size	Size, Brand Recognition	Size, Brand Recognition, CBA	Size	Size, Brand Recognition	Size, Brand Recognition, CBA
Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector
Fixed Effects						
	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector

Note: The outcome variable in Columns 4-6 is an indicator for whether the worker listed each of the firms in the researcher-provided not randomly assigned to them in the initial survey. Regressions use sampling weights. Standard errors are clustered at the worker level.

Randomization of Pay Offers to Firms

[▶ Back](#)

	Initial Survey	Follow-Up
	(1)	(1)
Number of Employees	0.39	0.31
<u>Sector</u>		
Manufacturing	0.64	0.64
Retail	0.44	0.09
Professional Services	0.71	0.15
Information Services	0.33	0.92
Transportation	0.21	0.34
Finance	0.74	0.27
<u>Other Firm Characteristics</u>		
HQ in Eastern Germany	0.78	0.59
Year of Incorporation	0.79	0.20
<u>Financial Characteristics</u>		
Total Assets per Employee	0.94	0.36
Fixed Assets per Employee	0.85	0.30

Note: We regress each covariate (indicated in the row) on the randomly assigned pay offer, controlling for the position of the firm (whether listed first, second, or third), and clustering standard errors at the worker level. Each entry is the p-value from a test that the coefficient on the pay offer is zero.

Robustness to Additional Specifications [▶ Back](#)

	Alternative Specifications of Distance			Alternative Samples		Alternative Weighting Schemes	
	Quadratic in Distance	Direct Distance	Closest Establishment	Initial Survey Only	Follow-Up Only	Unweighted	Population Weights
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Own-Pay Expectation	0.312*** (0.050)	0.309*** (0.050)	0.331*** (0.050)	0.427*** (0.122)	0.249 (0.174)	0.225*** (0.026)	0.255*** (0.085)
Observations	21272	21272	21272	15095	5980	21272	21236
Number of Workers (Clusters)	6440	6440	6440	0.396	0.309	0.330	0.335
Mean of Outcome	0.31	0.31	0.31	0.35	0.24	0.33	0.32
Implied Elasticity	1.004*** (0.162)	0.996*** (0.161)	1.067*** (0.161)	1.214*** (0.347)	1.052 (0.736)	0.677*** (0.079)	0.788*** (0.261)

What Workers Believe About High Wage Firms [▶ Back](#)

Suppose you are comparing job opportunities at two different companies: Company 1 pays 10% above the market average and Company 2 pays 30% above the market average.

1. Which company do you think attracts more qualified applicants per opening?
 - ▶ Company 1
 - ▶ Company 2
 - ▶ Both attract the same number of applicants

2. Which company do you think provides better non-wage amenities (e.g., home office, childcare subsidy)?
 - ▶ Company 1
 - ▶ Company 2
 - ▶ Both provide the same non-wage amenities

Reynolds (1951) [▶ Back](#)

*“These results confirm the prevalent impression that workers are poorly informed about job opportunities. Moreover, it is doubtful how far the situation can be altered by collecting and disseminating additional job information. **The basic difficulty is that satisfactorily employed workers are almost entirely uninterested in employment conditions in other companies. This lack of interest is an even more serious obstacle than the difficulty of compiling accurate job information [emphasis added]**”*

–Reynolds (1951)

What Could Information Do?

[▶ Back](#)

	High Attachment ("Provided Firms")		Low Attachment ("Consideration Set Firms")	
	Search	Mobility	Search	Mobility
	(1)	(2)	(3)	(4)
A. Below Median-Wage Workers				
<u>Full Transparency</u>				
Baseline $\lambda = 0.5$	12%	7%	12%	9%
High Compression $\lambda = 0.25$	19%	10%	19%	13%
Low Compression $\lambda = 0.75$	6%	3%	6%	4%
<u>Eliminate Attachment</u>				
Baseline $\lambda = 0.5$	34%	24%	14%	12%
B. All Workers				
<u>Full Transparency</u>				
Baseline $\lambda = 0.5$	8%	6%	7%	7%
High Compression $\lambda = 0.25$	12%	9%	10%	10%

Alternative Specifications for the Incumbent Premium [▶ Back](#)

	Alternative Specifications of Distance				Alternative Samples		Alternative Weighting	
	Log	Quadratic	Direct	Closest	Initial	Follow-Up	Population	
	Distance	in Distance	Distance	Establishment	Survey Only	Only	Unweighted	Weighted
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Researcher-Provided Firms								
Implied Incumbent Premium	0.112*** (0.019)	0.123*** (0.011)	0.113*** (0.019)	0.167*** (0.011)	0.142*** (0.025)	0.083*** (0.023)	0.127*** (0.007)	0.058*** (0.043)
Observations	29961	29961	29961	29961	16594	13367	29961	29918
Number of Workers	7735	7735	7735	7735	4322	2351	7735	7724
B. Worker-Provided Firms								
Implied Incumbent Premium	0.079*** (0.010)	0.078*** (0.008)	0.066*** (0.009)	0.072*** (0.008)	---	---	0.099*** (0.005)	0.053*** (0.012)
Observations	15259	15259	17539	17539	---	---	15259	15242
Number of Workers	4784	4784	4796	4796	---	---	4784	4778